List of Publications

The contents of this thesis have been published in the following articles:

- Section 1.5: Adapted from [7], where we solved a problem of Brian and Clontz.
- Chapter 2: Primarily based on my contributions to [3], specifically regarding the separation of learnability classes with computational structures, and the connections to other fields.
- Section 3.2: Focuses solely on my contributions to two main algorithms, their analysis, development and implementation detailed in [8].
- Section 3.3: Based on both the joint work and the sections and main results I contributed to [4] and [1].
- Section 3.4: Based on [5], specifically on my contributions to several conjectures and practical implementations.

Contents

List of Publications **i**

1 Descriptive Combinatorics **1**

1.1 Background . . . 1

1.2 A concise proof of the L_0 dichotomy . . . 1

1.2.1 Finite approximations . . . 1

1.2.2 A notion of smallness for copies . . . 3

1.2.3 Construction of a minimal graph of Borel chromatic number at least 3 . . . 5

1.2.4 Proof of Theorem 1.8 . . . 5

1.3 A Ramsey-type theorem for E_0 trees . . . 6

1.4 Inexistence of a basis for digraphs of dichromatic number 3 . . . 6

1.5 Uncountable sets and an infinite linear order game . . . 6

2 Deep Computations **7**

2.1 Introduction . . . 7

2.2 Connections to learning theory . . . 7

2.3 The Bourgain-Fremlin-Talagrand theorems . . . 7

2.4 The non-independence property . . . 7

2.5 Compositional computation structures . . . 7

2.6 Quantum logics . . . 7

3 Group Actions on Graphs **8**

3.1 Introduction . . . 8

3.2 Border tracing in arbitrary adjacency graphs . . . 8

3.3 Recursive constructions . . . 8

3.4 Amoeba trees . . . 8

3.5 Infinitary amoeba graphs . . . 8

Chapter 1

Descriptive Combinatorics

This is the descriptive combinatorics chapter of the thesis.

1.1 Background

This is the place to explain how to get from Cantor's classical results on Polish spaces, to the G_0 dichotomy and its applications, to more advanced topics.

1.2 A concise proof of the L_0 dichotomy

I present my own proof of the L_0 dichotomy. This approach simplifies the classical arguments and highlights the core combinatorial structures required for the subsequent theorems.

Throughout this entire work, G denotes an analytic graph on a Polish space $V(G) = X$. That is, X is a separable completely metrizable topological space, and G is a symmetric and irreflexive relation on X which, as a subspace of $X \times X$ is the continuous image of a Polish space.

I denote the Borel chromatic number of G by $\chi_B(G)$, that is, the least number of colours k needed to find a mapping $c : X \to k$ such a way that adjacent vertices in G are given different colours. I employ the symbol $H \to_c G$ (respectively, $H \to_B G$) to abbreviate the situation where there is a continuous (respectively Borel) homomorphism from the analytic graph H to G. It is straightforward to show the following.

Proposition 1.1. *If $H \to_c G$ or, more generally if $H \to_B G$, then $\chi_B(H) \leq \chi_B(G)$.*

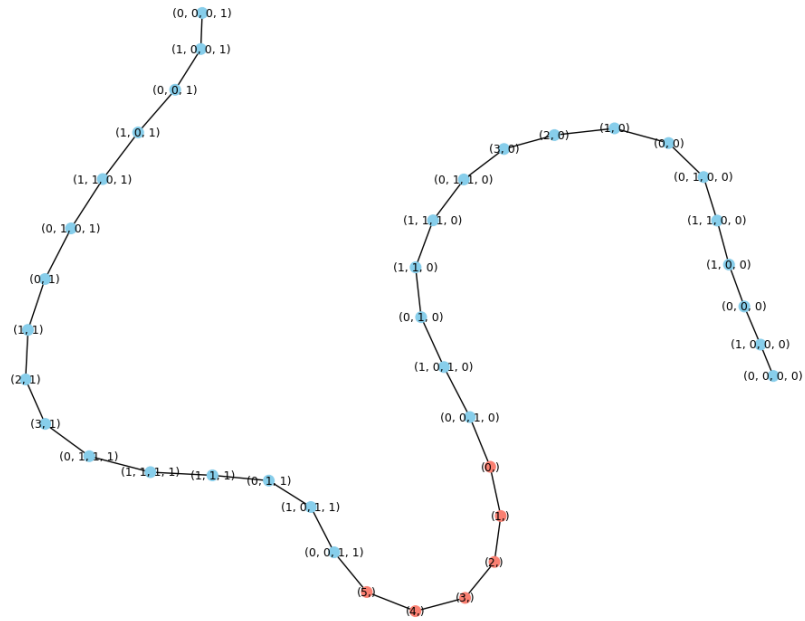
Using a standard greedy algorithm argument, it is clear that finite graphs G of maximum degree Δ satisfy $\chi(G) \leq \Delta + 1$. Interestingly, a similar fact holds for Borel graphs of uniformly bounded degree.

Theorem 1.2 (Proposition 4.6 in [6])**.** *If G is a Borel graph on a Polish space all of whose degrees are bounded by the natural number k, then $\chi_B(G) \leq k + 1$.*

1.2.1 Finite approximations

Fix the parameter $c \in \omega^\omega$ and a path of length ω on the vertex set $\{p_n\}_{n<\omega}$. That is, $\{(p_n, p_{n+1}) : n < \omega\}$. Naturally, I could assume that $p_n = n$, but this labeling will aid distinguish it from other

Figure 1.1: L_3^c where $c(n) = 2n - 1$.



indices.

I recursively construct a family of graphs $\{L_n^c : n < \omega\}$ such that for all $n < \omega$, the following properties hold. Take note of the fact that to construct up to L_n^c, I only require knowledge of $c \restriction n$.

(i) L_n^c is a finite path.

(ii) If $n > 0$, the endpoints of L_n^c are $e_i^n := (p_0)^\frown(0)^{n-1}{}^\frown(i)$, for $i < 2$.

(iii) Every vertex of L_n^c is of the form $(p_k)^\frown t$ for some k and $t \in 2^{\leq n}$. Unless $t = \emptyset$, I say the vertex is a *non-path* vertex.

Let $L_0^c = \{(p_0, \ldots, p_{c(0)})\}$ and suppose that L_n^c has been constructed. L_{n+1}^c is a path obtained from joining two copies of L_n^c by a path of length $c(n)$ (meaning, $c(n)$ *edges*), but I must update the vertex labels in order to distinguish the two copies. I do this by simply appending 0 to each vertex of the first copy, and 1 to the other. Formally, I start by adding the vertex $v^\frown(i)$ for every $i < 2$ and $v \in L_n^c$. Adding the path $((e_1^n)^\frown(0), p_0, p_1, \ldots, p_{c(n)}, (e_1^n)^\frown(1))$ completes the construction of L_{n+1}^c, and its endpoints are

$$e_i^{n+1} := (e_0^n)^\frown(i) = (p_0)^\frown(0)^n{}^\frown(i),$$

for $i < 2$. This finished the recursion.

Understanding the intuition behind what (iii) says about a vertex v in L_n^c, the number $|v| - 1$ indicates that v was added that many "recursive stages ago" at the position of p_k in the path joining the two copies of the previous paths; the sequence t tells the story, in order, of which of the two copies of the previous path the vertex lies in. In other words, vertices of the form $(p_k)^\frown t$ for $t \in 2^n$

are precisely those that were copied from the original L_0^c in all possible positions along L_n^c, doubling the number of copies at each stage, for example.

Based on our discussion, a vertex in some L_n^c is completely determined by the stage $n-m$ at which it was added in the position of p_k and then copied according to t.

1.2.2 A notion of smallness for copies

By definition, the set of edges of G is the continuous image of some Polish space E, let us say under π. For a finite graph H, consider $\mathrm{Hom}(H,G)$ the set of all maps

$$\varphi : V(H) \to V(G)$$
$$\varphi : E(H) \to E$$

that satisfy that for all $uv \in E(H)$, $\pi(\varphi(uv)) = \varphi(u)\varphi(v)$. Formally, φ is really two maps, however I assume that $V(H) \cap E(H) = \emptyset$ and so there is never any confusion. Notice that $\mathrm{Hom}(H,G)$ is a Polish space, being a closed subset of the product space $X^{V(H)} \times (E)^{E(H)}$.

Given $\mathcal{H} \subseteq \mathrm{Hom}(H,G)$, I define $\mathcal{H}(u) := \{\varphi(u) : \varphi \in \mathcal{H}\}$, which is analytic whenever $\mathcal{H}$ is Borel. Also, $\mathcal{H}(uv) := \{\varphi(uv) : \varphi \in \mathcal{H}\}$.

Definition 1.3.

1. Let $k < \omega$. I say an analytic set $A \subseteq V(G)$ has property $\Phi(A,k)$ if all odd walks of G with endpoints in A have length at most $2k-1$.

2. For, $n > 0$, a Borel set $\mathcal{L} \subseteq \mathrm{Hom}(L_n^c, G)$ is called *tiny* if there is a non-path vertex $u \in V(L_n^c)$ and a natural k such that $\Phi(\mathcal{L}(u), k)$. For $n = 0$, L_n^c is just an edge consisting of the two path vertices p_0 and p_1, so in this case I define $\mathcal{L} \subseteq \mathrm{Hom}(L_0^c, G)$ as *tiny* if there is a natural k such that $\Phi(\mathcal{L}(p_1), k)$.

3. A set $\mathcal{L} \subseteq \mathrm{Hom}(L,G)$ is *small* if it is the countable union of tiny sets. This notion defines a σ-ideal.

4. A set is *large* if it is not small. Notice that when H is the graph of one single vertex, I can define $\mathrm{Hom}(H,G)$ to be $V(G)$ and then for any $\mathcal{H} \subseteq \mathrm{Hom}(\cdot, G)$, $\mathcal{H}(\cdot)$ is essentially the same as $\mathcal{H}$.

I need the following auxiliary result that links the 2-colourability and countable chromaticity of a graph in this context.

Lemma 1.4 (Claim 3.5 of [2])**.** *If $\Phi(A,n)$, then there is an invariant (that is, union of connected components) Borel set $B \supseteq [A]_{E_G}$ that induces a 2-Borel-colourable subgraph of G, that is, there is a Borel proper colouring $c_B : G \restriction B \to 2$.*

Theorem 1.5. *If $\chi_B(G) > 2$ then $V(G)$ is large.*

Proof of theorem. Suppose that $V(G) = \bigcup_{m<\omega} \mathcal{H}_m$ where each $\mathcal{H}_m$ is a tiny Borel set. By Lemma 1.4, there is a G-invariant Borel set $B_m \supseteq [\mathcal{H}_m]_{E_G}$ with a Borel 2-coloring c_m. Define $c : V(G) \to 2$ by letting $m(x) := \min\{m < \omega : x \in B_m\}$ and $c(x) = c_{m(x)}(x)$. Then c is a Borel 2-colouring because each B_m is G-invariant. $\square$

Since L^c_{n+1} is obtained from two copies of L^c_n joined by a path, I require a way to refer to the distinct copies of L^c_n inside of L^c_{n+1}. By (iii) above, for every $\varphi \in \mathrm{Hom}(L^c_{n+1}, G)$, I define $\varphi^i \in \mathrm{Hom}(L^c_n, G)$ on every non-path vertex u of L^c_n by $\varphi^i(u) = \varphi(u^\frown(i))$ for all $u \in V(L^c_n)$ and $\varphi^i(uv) = \varphi(u^\frown(i), v^\frown(i))$ for all $(u, v) \in E(L^c_n)$. Thus, given $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$, define $\mathcal{L}^{+c}$ as the set of all $\varphi \in \mathrm{Hom}(L^c_{n+1}, G)$ for which $\varphi^i \in \mathcal{L}$ for both $i < 2$. Notice that if $\mathcal{L}$ is Borel, then so is $\mathcal{L}^{+c}$.

The length of the path I add needs to be updated dynamically throughout the proof. Formally, I may need to adjust the parameter c by moving into a different branch of the Baire space when constructing the graphs L^c_n. Notice that, for two reals $c, d \in \omega^\omega$, if $d \restriction n = c \restriction n$, then $L^c_k = L^d_k$ for all $k < n$.

Lemma 1.6. *Let $n > 0$.*

(a) Suppose that c only takes odd values, that k is a natural and $t \in 2^{<n}$ is such that $(p_k)^\frown t^\frown(i)$, for $i < 2$, are non-path vertices of L^c_n (see (iii) above). Then, they are an odd distance apart in L^c_n.

(b) If $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d \restriction n = c \restriction n$ and $\mathcal{L}^{+d}$ is a nonempty subset of $\mathrm{Hom}(L^d_{n+1}, G)$.

(c) If $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d \restriction n = c \restriction n$ and $\mathcal{L}^{+d}$ is a large subset of $\mathrm{Hom}(L^d_{n+1}, G)$.

Proof. For (a), observe that both vertices come from the same vertex in L^c_{n-1}, and so their distance in L^c_n is twice the distance to e^{n-1}_1 plus the length of the joining path, hence odd.

(b) follows from the fact that if $\mathcal{L}$ is large, it is not tiny. Hence, $\mathcal{L}(e^n_1)$ contains the endpoints of a minimal odd path of length $k \geq 2n + 1 > 2$, that is, there are $\varphi_0, \varphi_1 \in \mathcal{L}$ and a path in G ($\varphi_0(e^n_1) = v_0, \ldots, v_k = \varphi_1(e^n_1)$). Let $e_1, \ldots, e_k \in E$ be such that $\pi(e_j) = (v_{j-1}, v_j)$. Defining $d(n) = k-2$ and $d(x) = c(x)$ for all $x \neq n$, notice that, by the remark preceding this lemma, $L^c_n = L^d_n$. I find $\varphi \in \mathrm{Hom}(L^d_{n+1}, G)$ be given by $\varphi^0 = \varphi_0$, $\varphi^1 = \varphi_1$ and $\varphi(p_i) = v_{i+1}$ for $0 \leq i \leq k-2 = d(n)$; the edges are given by $\varphi(p_j, p_{j+1}) = e_j$ for $0 \leq j < k-2$, $\varphi(e^n_1{}^\frown(0), p_0) = e_1$ and $\varphi(p_{k-2}, e^n_1{}^\frown(1)) = e_k$.

I prove (c) by contradiction. Recall that constructing L^c_{n+1} from L^c_n requires only knowing the value of $c(n)$. Suppose that for all d that are equal to c except in the n-th value, $\mathcal{L}^{+d} = \bigcup_m \mathcal{F}^d_m$ where each $\mathcal{F}^d_m$ is a tiny Borel subset of $\mathrm{Hom}(L^d_{n+1}, G)$. For each m and d, let $v^d_m \in V(L^d_n)$, $i^d_m < 2$ and $n_d(m)$ be the minimal natural such that $\Phi(\mathcal{F}^d_m(v^d_m{}^\frown(i^d_m)), m_d(n))$.

Now use the Reflection Lemma (reference **??** to be added) to find a Borel $B^d_m \supseteq \mathcal{F}^d_m(v_m^\frown(i_m))$ such that $\Phi(B^d_m, m_d(n))$. Consider $\mathcal{H}^d_m = \{\varphi \in \mathcal{L} : \varphi(v^d_m) \in B^d_m\}$, which is a Borel tiny subset of $\mathrm{Hom}(L^d_n, G) = \mathrm{Hom}(L^c_n, G)$. Indeed, notice that $\mathcal{H}^d_m(v^d_m) = \{\varphi(v^d_m) : \varphi \in \mathcal{H}^d_m\} = \{\varphi(v^d_m) : \varphi \in \mathcal{L} \wedge \varphi(v^d_m) \in B^d_m\} \subseteq B^d_m$.

Consider the large Borel set $\mathcal{L}_-$ obtained from removing from $\mathcal{L}$ the countable union of all tiny sets of the form $\mathcal{H}^d_m$. By part (b), there exist φ and d such that $\varphi \in \mathcal{L}^{+d}_- \subseteq \mathcal{L}^{+d} \subseteq \mathrm{Hom}(L^d_{n+1}, G)$. Suppose that $\varphi \in \mathcal{F}^d_m$, then $\varphi^{i^d_m}(v_m) = \varphi(v^d_m{}^\frown(i^d_m)) \in \mathcal{F}^d_m(v^d_m{}^\frown(i^d_m)) \subseteq B_m$, which means that $\varphi^{i^d_m} \in \mathcal{H}^d_m \cap \mathcal{L}_- = \emptyset$; a contradiction. $\square$

1.2.3 Construction of a minimal graph of Borel chromatic number at least 3

The goal of this sections it to construct a family of graphs $\mathbb{L}_c$ indexed by reals $c \in \omega^\omega$.

Now, fix c and consider X_c as the set of all tuples $(n, k, x) \in \mathbb{N} \times \mathbb{N} \times 2^{\mathbb{N}}$ with $k \leq \max\{1, c(n)\}$ and notice that it is a closed subspace of the product space, and hence Polish. Define, for each (m, k, x) with $m \leq n$, $\pi_n(m, k, x)$ as the vertex of L_n^c determined by these parameters. Formally, $\pi_n(m, k, x) := (p_k)^\frown x \restriction (n - m)$. For example, $\pi_3(1, 2, 0110\cdots) = (p_2, 0)$ can be seen in Figure 1.1 labeled as $(2, 0)$.

Finally, $\mathbb{L}_c$ is the graph on X_c where two (n_i, k_i, x_i), for $i < 2$, are adjacent if and only if the $\pi_n(n_i, k_i, x_i)$ are adjacent in all L_n^c with $n \geq \max\{n_0, n_1\}$. Some basic properties of $\mathbb{L}_c$ follow.

Lemma 1.7.

(a) Two vertices $(n_i, k_i, x_i) \in X_c$, for $i < 2$, are in the same connected component of $\mathbb{L}_c$ if and only if there are $t_i \in 2^{<\omega}$ and $x \in 2^\omega$ such that $|t_0| - |t_1| = n_1 - n_0$ and $x_i = t_i^\frown x$.

(b) $\mathbb{L}_c$ has no cycles and all vertices of $\mathbb{L}_c$ have degree 2 except for the vertex $(0, 0, \{0\}^\omega)$, which has degree 1.

(c) $\chi_B(\mathbb{L}_c) = 3$.

Proof. (c) That $\chi_B(\mathbb{L}_c) \leq 3$ follows from Proposition 1.2 and the preceding proposition. I argue the other inequality by contradiction. Suppose then that there is a partition of X into two independent sets. Since X is Polish, there is a non-meager class B in the partition. B is comeager in some basic open set $[(n, k, t)]$. Skipping some technical details, this means that I can find a vertex $(n, k, t^\frown(\varepsilon)^\frown x) \in B$, and moreover not connected to any member of $[(n, k, t)] \setminus B$. I claim that $(n, k, t^\frown(1 - \varepsilon)^\frown x) \in B$.

It then follows from Lemma 1.6(a) that these two vertices in B are an odd distance apart, which is impossible. □

The main result is then split into two parts.

Theorem 1.8. *For any analytic graph G, either $\chi_B(G) \leq 2$ or there is a c such that $\mathbb{L}_c \to_c G$.*

Moreover, c can be chosen to be unbounded and take only odd values.

Theorem 1.9. *Let $c, d \in \omega^\omega$ be unbounded and only taking odd values, then $\mathbb{L}_c$ and $\mathbb{L}_d$ are continuously homomorphically equivalent.*

1.2.4 Proof of Theorem 1.8

By Lemma 1.7(c) and Proposition 1.2, it is clear that both conditions cannot hold simultaneously. Now suppose that G is an analytic graph of Borel chromatic number at least three. The theorem is proved by constructing a valid c and a continuous homomorphism witnessing $\mathbb{L}_c \to_c G$.

By Theorem 1.5, $V(G)$ is large. I recursively construct a function $c \in \omega^\omega$ and a sequence of large Borel sets $\mathcal{L}_n \subseteq \mathrm{Hom}(L_n^c, G)$ such that for all n,

1. $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$;

2. for all $u \in V(L_n^c)$, the diameter of $\overline{\mathcal{L}_n(u)}$ is less than 2^{-n}; and

3. for all $(u, v) \in E(L_n^c)$, the diameter of $\overline{\mathcal{L}_n(u, v)}$ is less than 2^{-n}.

By (1), $\mathcal{L}_{n+1}(\pi_{n+1}(m+2, k, x)) \subseteq \mathcal{L}_n(\pi_n(m, k, x))$ for all $n > m$ and similarly for the edges: For $(m, k, x) \in X_c$, let $f(m, k, x)$ be the unique point in

$$\bigcap_{n=m}^{\infty} \overline{\mathcal{L}_n(\pi_n(m, k, x))} = \bigcap_{n=0}^{\infty} \overline{\mathcal{L}_{n+m}((p_k)^\frown x \restriction n)}.$$

The map $f : X_c \to X$ is continuous.

It remains to check that it is a homomorphism from $\mathbb{L}_c$ to G. Suppose that $(m_i, k_i, x_i)_{i<2}$ are adjacent in $\mathbb{L}_0$, that is, for all $n \geq n_0 := \max\{m_0, m_1\}$, $\pi_n(m_i, k_i, x_i)_{i<2} \in L_n^c$. There is a unique edge $e \in E$ inside

$$\bigcap_{n \geq n_0} \overline{\mathcal{L}_n(\pi_n(m_0, k_0, x_0), \pi_n(m_1, k_1, x_1))}.$$

By continuity, $(f(m_0, k_0, x_0), f(m_1, k_1, x_1)) = \pi(e) \in G$. Thus the images form an edge of G.

1.3 A Ramsey-type theorem for E_0 trees

I present a Ramsey-type theorem for E_0 trees, which is a key combinatorial ingredient in ongoing work.

1.4 Inexistence of a basis for digraphs of dichromatic number 3

I answer the open question of whether a basis exists for the class of digraphs with dichromatic number at least 3.

1.5 Uncountable sets and an infinite linear order game

This section addresses a question posed by Matthew Baker [Math. Mag. 80 (2007)] regarding the characterization of countable subsets of the reals via infinite games. While Brian and Clontz recently characterized this for countable payoff sets, the question remained open for general linear orders.

My Contribution: In collaboration with the author of [7], I provided the positive answer to the existence of winning strategies on uncountable sets. Furthermore, we constructed a dense linear order of size κ for every infinite cardinal κ on which Player II has a winning strategy. This construction demonstrates the complexity of generalizing the Brian-Clontz characterization.

Chapter 2

Deep Computations

This is the deep computations chapter of the thesis.

2.1 Introduction

2.2 Connections to learning theory

I formalize the connection between deep computations and learning theory. Expanding on the initial concepts from [3], I explicitly map the relationships between the Non-Independence Property (NIP), VC-dimension, and PAC-learnability within this new context.

2.3 The Bourgain-Fremlin-Talagrand theorems

I synthesize the Bourgain-Fremlin-Talagrand theorems, organizing and refining the material presented during four seminars at the Fields Institute into a cohesive reference for this field.

2.4 The non-independence property

I analyze the model-theoretic aspects of deep computations, isolating the specific role NIP plays in the combinatorial properties of definable sets.

2.5 Compositional computation structures

This section details my primary contribution to [3]: the construction of concrete examples of computation structures that separate learnability classes. These examples provide the necessary counterpoints to establish the boundaries of the theory.

2.6 Quantum logics

I present a theoretical framework that connects deep computations to quantum logics.

Chapter 3

Group Actions on Graphs

This is the group actions on graphs chapter of the thesis.

3.1 Introduction

3.2 Border tracing in arbitrary adjacency graphs

While border tracing is standard for finite planar graphs, I generalize these algorithms to arbitrary adjacency graphs.

My Contribution: I developed and analyzed two generalized algorithms for this setting.

3.3 Recursive constructions

I summarize my contributions to both [4] and [1] regarding recursive constructions of graphs with interesting combinatorial properties.

3.4 Amoeba trees

I present the properties of amoeba trees and recent advancements regarding their conjectures.

My Contribution: I independently conceptualized several key properties discussed in this section. Additionally, I authored abstractions and object-oriented frameworks that had a significant impact on the results presented in [5].

3.5 Infinitary amoeba graphs

I synthesize ideas on infinite graphs, focusing on the interaction between automorphism groups, ends, and amenable groups, specifically placing my own results and ideas concerning amoeba graphs within this broader theoretical context.

Definition 3.1 (Local Amoeba)**.** A graph G is called a *local amoeba* if $\overline{\mathrm{Fer}(G)} = S_X$.

Assume that G is a graph defined on an infinite countable set X. In this case, X^X is homeomorphic with the Baire space ω^ω, and so we are dealing with separable and completely metrizable spaces (i.e. Polish spaces). One can show that S_X, being a G_δ subspace of the Baire space, is also Polish and thus its topology is induced by a complete metric. Indeed, observe that

$$S_X = \bigcap_{\substack{x,y\in X\\ x\neq y}} \{f \in X^X : f(n) \neq f(m)\} \cap \bigcap_{y\in X} \bigcup_{x\in X} \{f \in X^X : f(x) = y\}$$

is a countable (whenever X is) intersection of open sets. Giving an explicit complete metric is done in the example at the end of the section.

Group actions on graphs may be quite different in the infinite case. For instance, consider the automorphism of a random graph. Most finite random graphs are asymmetric but almost all countably infinite random graphs are highly symmetric.

Many properties of finite amoebas translate to our more general definition, such as the fact that $\mathrm{Fer}(G) = \mathrm{Fer}(\overline{G})$ for any graph G. Also, any complete graph on any set X is a local amoeba, and thus so is the discrete graph (the graph with no edges). As a first non-trivial example, the bi-infinite path $P_\mathbb{Z}$, i.e. the graph defined on $X = \mathbb{Z}$ where the edges are given by consecutive pairs, is interestingly not a local amoeba. In fact in this context, bi-infinite paths play a similar role as finite cycles. Indeed, it straightforward to verify that $\mathrm{Fer}(P_\mathbb{Z}) = \mathrm{Aut}(P_\mathbb{Z})$. Notice that the basic open set $[(0\ 1)(2)]$ is disjoint from $\mathrm{Fer}(G)$ since the only automorphism interchanging 0 and 1 is a reflection (which doesn't fix the element 2), and thus $(0\ 1) \notin \overline{\mathrm{Fer}(G)}$.

Lemma 3.2. *Let $\Gamma \leq S_X$ and $\varphi \in S_X$. Then $\varphi \in \overline{\Gamma}$ if and only if for every finite $F \subseteq X$, there is $\sigma \in \Gamma$ such that $\varphi \restriction F = \sigma \restriction F$.*

Proof. For the forward direction, take φ and F as above. Clearly, $[\varphi \restriction F]$ is a basic open neighborhood of φ and hence must intersect Γ. Any σ witnessing this has the sought property.

Conversely, take an arbitrary $t \in \omega^{<\omega}$ extended by φ. Pick $\sigma \in \Gamma$ with $\varphi \restriction \mathrm{dom}\, t = \sigma \restriction \mathrm{dom}\, t$. Clearly, $\sigma \in \Gamma \cap [t]$, proving that $\varphi \in \overline{\Gamma}$. □

A group action on X is called *n-transitive* if $|X| \geq n$ and for any two pairwise distinct n-tuples $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$, there is a group element g such that $gx_i = y_i$ for all $1 \leq i \leq n$. Notice that 1-transitive is just the usual definition of *transitive*, i.e. that there is a single orbit. If for all $n \geq 1$, $\mathrm{Fer}(G)$ acts n-transitively on X, we say $\mathrm{Fer}(G)$ is *highly transitive*.

Proposition 3.3. *The following are equivalent.*

1. *The graph G is a local amoeba.*
2. *Every finite $F \subseteq X$ and every $\varphi \in S_X$, there is $\sigma \in \mathrm{Fer}(G)$ such that $\varphi \restriction F = \sigma \restriction F$.*
3. *$\mathrm{Fer}(G)$ is highly transitive.*

Proof. The implication $[1 \implies 2]$ follows from the previous lemma. Assume (2) and let $n \geq 1$ and $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$ be two pairwise distinct. Let $F := \{x_i : 1 \leq i \leq n\}$, then fix a bijection $\omega \setminus F \to \omega \setminus \{y_i : 1 \leq i \leq n\}$ and define φ such that it agrees with said bijection and in addition satisfies $\varphi(x_i) = y_i$. By (2), some $\sigma \in \mathrm{Fer}(G)$ has the desired property.

Finally, assume (3), let φ be an arbitrary permutation on X and $F = \{x_1, \ldots, x_n\} \subseteq X$. Apply (3) to the tuples $(x_1, \ldots, x_n)$ and $(\varphi(x_1), \ldots, \varphi(x_n))$ to get $\sigma \in \mathrm{Fer}(G)$. It follows that σ agrees with φ on F. Since F was arbitrary, by the previous lemma, G must be a local amoeba. □

I point out that in the example of the bi-infinite path, $\mathrm{Fer}(P_{\mathbb{Z}})$ acts transitively on $P_{\mathbb{Z}}$ but not 2-transitively, as the same aforementioned example clearly shows, namely the pairs $(0, 2)$ and $(1, 2)$. In such cases, one could use the largest n for which the group acts n-transitively on G to compare how close G is to being a local amoeba.

Denote by P_{n+1} the path on the vertex set $n+1$ with edges given by consecutive pairs of numbers. For a label $i \in X$, denote by $\mathrm{Fer}^i(G)$ the subgroup of $\mathrm{Fer}(G)$ generated by those generators of $\mathrm{Fer}(G)$ that fix i. We say a graph G is *stem-symmetric with respect to* i (or to the vertex labeled by i) if $\mathrm{Fer}^i(G)$ is the symmetric group on $X \setminus \{i\}$.

The next result provides an example of a graph where the gap between the automorphism group and the Fer group is as large as possible. Concretely, the countably infinite path on $X = \omega$, denoted P_ω where the edges are given by consecutive pairs of naturals, has a trivial automorphism group, but $\mathrm{Fer}(P_\omega)$ is dense in S_ω.

We need the following combinatorial lemma to prove that infinite paths are local amoebas. The only need the fact that finite paths are stem-symmetric with respect to an endpoint.

Lemma 3.4. *The graph P_{n+1} is stem-symmetric with respect to k if and only if $k \leq n$ and, if $n \geq 5$ is odd, $k \neq \frac{n-1}{2}$.*

Proof. We prove the case $k = n$ as the others are similar. Let $1 \leq \ell < n$ and notice that the graph $P_{n+1} - (\ell, \ell+1) + (0, \ell+1)$ is a path on $n+1$ vertices and thus isomorphic to P_{n+1}. We can find an explicit isomorphism that fixes n and decompose it into transpositions. In symbols,

$$\prod_{k=0}^{\ell} (k \quad \ell - k) : (\ell \ (\ell+1) \to 0 \ (\ell+1)).$$

The second step is realizing that these permutations indeed generate the symmetric group on (the set) n. This is straightforward and can be done in many ways. For example, every transposition of the form $(0 \ \ell)$, for $1 \leq \ell < n$, is in $\mathrm{Fer}^n(P_{n+1})$ by an inductive argument on ℓ, taking conjugates and noticing that the transposition in question is the left-most factor in the ℓ-th permutation above. □

Proposition 3.5. *P_ω is a local amoeba.*

Proof. Let $F \subseteq \omega$ be finite and $\varphi \in S_\omega$. Define $n := \max(F \cup \varphi[F]) + 1$ and notice that $t := \varphi \restriction F$ is a bijection between finite subsets of n. In particular, there exists a bijection $s : n \setminus F \to n \setminus t[F]$. Now define $\sigma : X \to X$ by

$$\sigma(x) := \begin{cases} s(x) & x \in n \setminus F \\ t(x) & x \in F \\ x & x \geq n \end{cases}$$

Routine arguments show that $\sigma \in S_X$. Now consider the induced subgraph P_{n+1} and the permutation $\sigma \restriction (n+1)$. Observe that this permutation fixes the endpoint of the path, n, by definition. Thus by the previous lemma, there is a sequence $f_0, \ldots, f_m$ of generators of $\mathrm{Fer}(P_{n+1})$, all of which fix n, such that $\sigma \restriction (n+1) = f_0 \cdots f_m$.

For each $i < m$, we can extend f_i to a permutation $\overline{f_i}$ on X by fixing every label $x > n$. Clearly, each $\overline{f_i}$ is a member of $\mathrm{Fer}(P_\omega)$ and thus $\sigma = \overline{f_0} \cdots \overline{f_m} \in \mathrm{Fer}(P_\omega)$. Finally, the choice of σ makes it evident that $\sigma \restriction F = t = \varphi \restriction F$, and thus P_ω is a local amoeba by Proposition 3.3. $\square$

Bibliography

[1] Y. Caro, I. González, A. Hansberg, M. Jácome, T. Matos-Wiederhold, and A. Montejano. Graphs with constant balancing number. *Procedia Computer Science*, 223:325–333, 2023. XII Latin-American Algorithms, Graphs and Optimization Symposium (LAGOS 2023). URL: `https://www.sciencedirect.com/science/article/pii/S1877050923010177`, `doi:10.1016/j.procs.2023.08.244`.

[2] Raphaël Carroy, Benjamin D. Miller, David Schrittesser, and Zoltán Vidnyánszky. Minimal definable graphs of definable chromatic number at least three.

[3] Eduardo Dueñez, José Iovino, Tonatiuh Matos-Wiederhold, Luciano Salvetti, and Franklin D. Tall. Complexity of deep computations via topology of function spaces, 2026. Preprint. `arXiv:2601.00528`.

[4] L. Eslava, A. Hansberg, T. Matos-Wiederhold, and D. Ventura. New recursive constructions of amoebas and their balancing number. *Aequationes Mathematicae*, 99:1265–1299, 2025. Preprint available at `https://https://arxiv.org/abs/2311.17182`.

[5] S. Gómez, I. González, A. Hansberg, and T. Matos-Wiederhold. Amoeba trees. Preprint.

[6] A.S. Kechris, S. Solecki, and S. Todorcevic. Borel chromatic numbers. *Advances in Mathematics*, 141(1):1–44, 1999.

[7] Tonatiuh Matos-Wiederhold and Luciano Salvetti. Uncountable sets and an infinite linear order game. *Topology Proceedings*, 66:177–184, 2025. E-published on July 22, 2025. Preprint available at `https://arxiv.org/abs/2408.14624`.

[8] Petra Wiederhold and Tonatiuh Matos-Wiederhold. Border tracing in oriented adjacency graphs of polygonal tilings. *Theoretical Computer Science*, 1065:115729, 2026. URL: `https://www.sciencedirect.com/science/article/pii/S0304397525006668`, `doi:10.1016/j.tcs.2025.115729`.